

A-MES

AUTOMOTIVE MES • SEOYON E-HWA

V O L 0 0

Project Overview & Data Foundation

Index • ERD • DB Schema

Document • L3 Detailed Design Specification

Generated • 05-26-2026 • Build 20260526-190315

Reference • Plant A " SAV Detroit, MI | Plant B " GEO Birmingham, AL

CONFIDENTIAL • FOR INTERNAL USE ONLY

TABLE OF CONTENTS

Project Overview & Data Foundation

INDEX	A-MES UI Design Specification — Index
-------	---------------------------------------

AMES_ERD	A-MES ERD
----------	-----------

AMES_DB_Schema	A-MES Database Schema
----------------	-----------------------

● AUTOMOTIVE MES · UI DESIGN SPECIFICATION

A-MES

MANUFACTURING EXECUTION SYSTEM

자동차 부품 제조실행시스템 UI 설계서

VERSION

3.1

CLIENT

서연이화

PLANT

SAV / GEO

VOLUMES

VOL.03 – VOL.13

11

MODULES / 모듈

2

BOP ROUTINGS

3

QC GATES

10

REPORT TYPES

29

MASTER DATA
ENTITIES

7

RBAC ROLES

150+

BUSINESS RULES

 PDF DELIVERABLES PDF 산출물

DOWNLOAD AS PDF

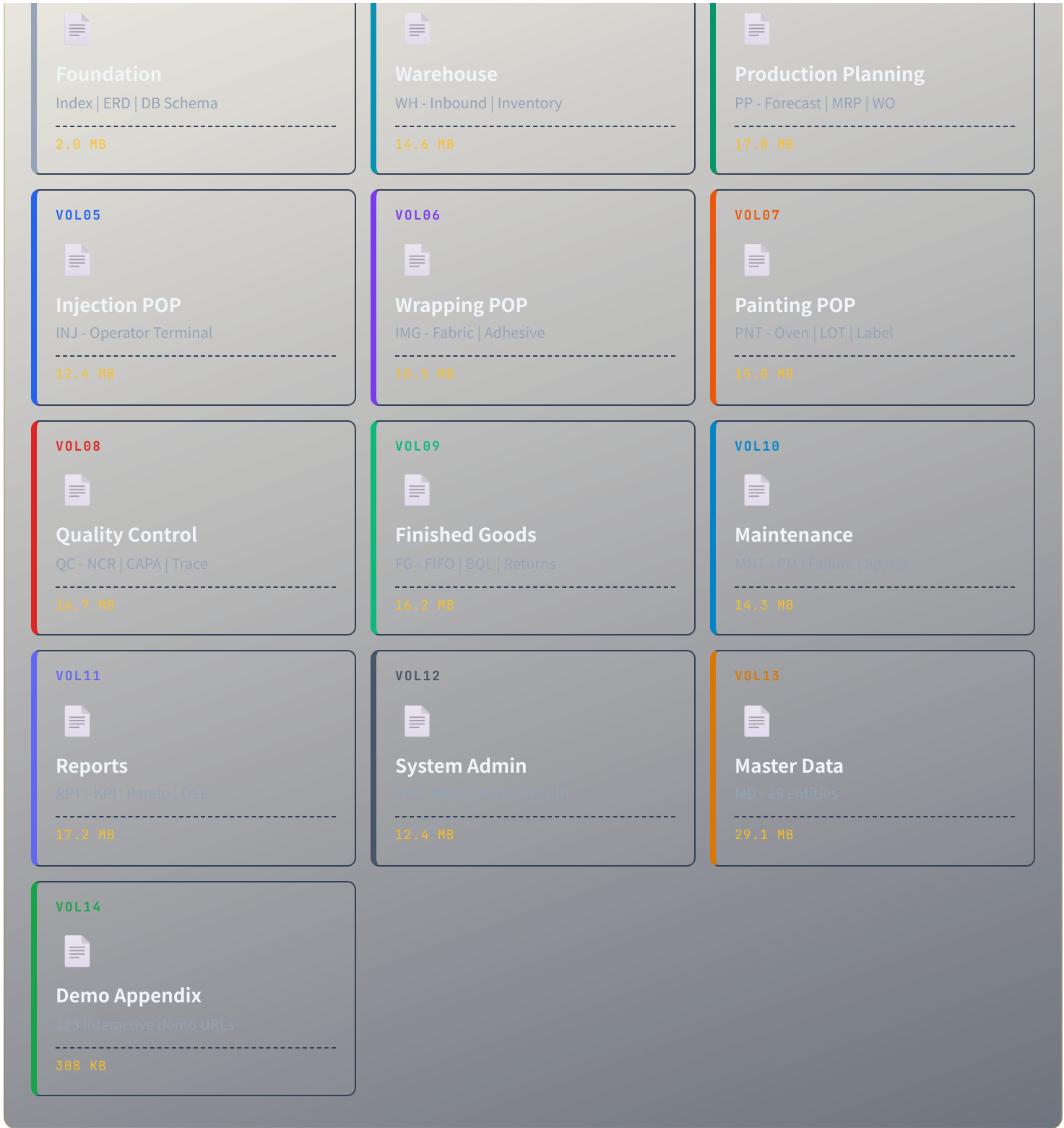
U.S. Letter | Auto-generated

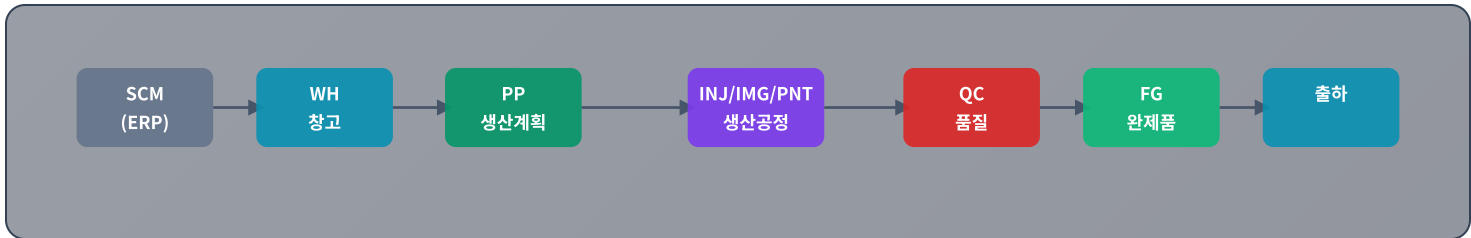
Each volume is available as a self-contained PDF (cover | TOC | all L3 specs). Click any tile below to download. Files are generated from the latest source HTML - rebuild via `.#tools#build_pdf.ps1`

VOL00

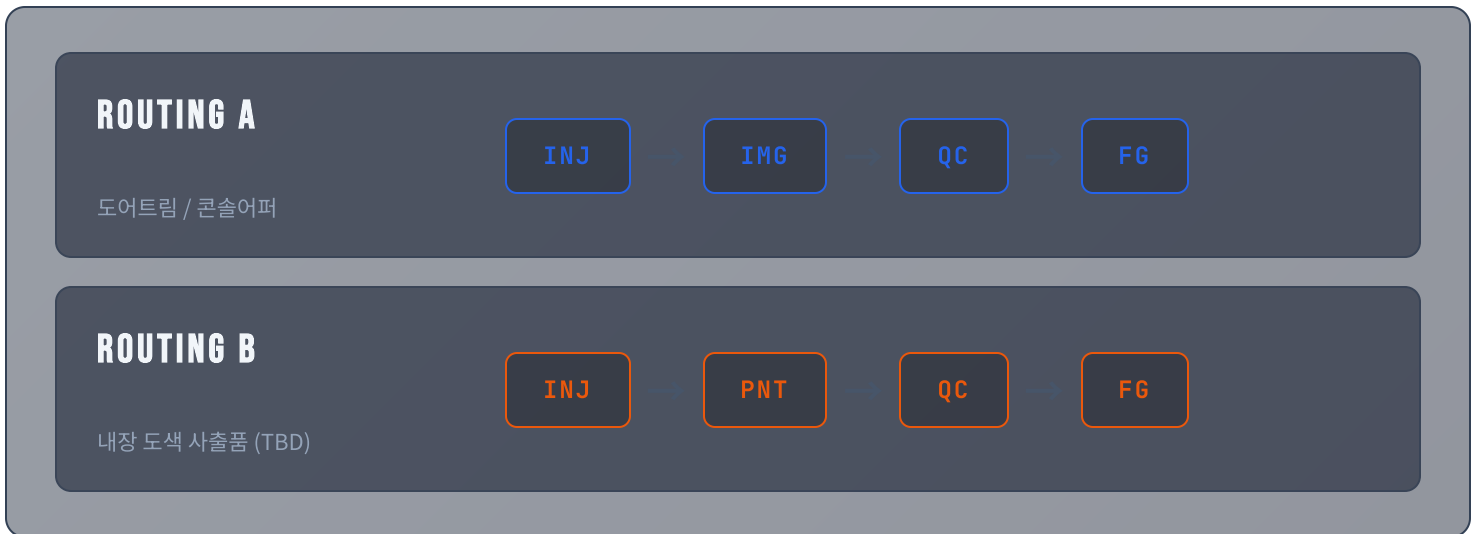
VOL03

VOL04





BOP PRODUCT ROUTING 제품 라우팅



MODULE INDEX 모듈 목록

VOL03 WH Warehouse Mgmt 창고관리 Receiving / PDA / Inventory / Location 입고·PDA·재고·로케이션 관리	VOL04 PP Production Plan 생산계획 Work Order / MRP / Line Schedule / OEE 작업지시·MRP·라인스케줄·가동/OEE
VOL05 INJ Injection 사출생산 POP / Defect / Mold Shot Count	VOL06 IMG Wrapping 감싸기생산 Fabric Roll / PDA 4-Step / Defect

POP·불량·금형 쇼트수 관리

원단롤·PDA 4단계·불량

VOL07

PNT

Painting

도장생산

5-Step Process / Oven Monitor / Defect

5단계공정·건조로모니터·불량

VOL08

QC

Quality Control

품질관리

3-Gate Inspection / Enhanced Flag

3게이트검사·강화검사 플래그

VOL09

FG

Finished Goods

완제품출고

FIFO Picking / Delivery / Shipping

FIFO피킹·거래명세서·출고

VOL10

MNT

Maintenance

설비관리

OEE / PM Calendar / Mold Block

OEE·PM달력·금형 강제차단

VOL11

RPT

Reports

분석리포트

10 Reports / Report Center / Builder

10종리포트·리포트센터·빌더

VOL12

SYS

System Admin

시스템관리

RBAC 7-Role / Config / Health / Audit

7역할RBAC·환경설정·헬스·감사로그

VOL13

MD

Master Data

기준정보

29 Entities / 5 Categories / Table Design

29개 엔티티·5카테고리·테이블설계

VOL03 · WH

Warehouse 창고관리

8 screens · L2 →

WH-01

Inbound Schedule

WH-02

PDA Inbound

WH-03

Inventory Status

WH-04

Location Map

WH-05

Inventory Adjust

WH-06

Release Schedule

WH-07

PDA Release

WH-08

Transaction History

VOL04 · PP

Production Plan 생산계획

13 screens · L2 →

PP-01

Forecast Mgmt

PP-02

SAP B1 Import

PP-03

Plan & Confirm

PP-04

Work Order Mgmt

PP-05

MRP

PP-06

Purchase Req.

PP-07

WO Release

PP-CAL

Production Calendar

PP-OTD

On-Time Delivery

PP-LSB

Line Schedule Board

PP-ODM

Downtime Monitor

PP-OEE

Line Availability & OEE

PP-DTL

Downtime Log Entry

VOL05 · INJ

Injection 사출생산

8 screens · L2 →

INJ-01

POP Login

INJ-02

POP Dashboard

INJ-03

WO Confirmation

INJ-04

Production Entry

INJ-05

Defect Register

INJ-06

Mold Change

INJ-07

Production Status

INJ-08

Andon Emergency

VOL06 · IMG

Wrapping 감싸기생산

7 screens · L2 →

IMG-01

POP Login

IMG-02

POP Dashboard

IMG-03

Production Entry

IMG-04

Fabric Input & Verify

IMG-05

Defect Register

IMG-06

Bond Setup

IMG-07

Production Status

VOL07 · PNT

Painting 도장생산

9 screens · L2 →

PNT-01

Daily Production Plan

PNT-02

Virtual LOT Pre-Issue

PNT-03

Loading Station (R1)

PNT-04

Line Tracking Board

PNT-05

Oven Monitor (R2)

PNT-06

Unloading & Count (R3)

PNT-07

LOT Label Apply

PNT-08

Defect Register

PNT-09

Shift Log / Report

VOL08 · QC

Quality Control 품질관리

9 screens · L2 →

QC-01

Incoming Inspection

QC-02

In-Process Inspection

QC-03

Final Inspection

QC-04

NCR Register

QC-05

Hold / Quarantine

QC-06

CAPA Management

QC-07

QC Dashboard

QC-08

Inspection Standard

QC-TRC

Traceability

VOL09 · FG

Finished Goods 완제품출고

10 screens · L2 →

FG-01

PDA Stocking

FG-02

FG Inventory

FG-03

Shipment Order

FG-04

PDA FIFO Picking

FG-05

Loading Confirmation

FG-06

Delivery Note

FG-07

Day-End Close

FG-08

Shipment History

FG-09

FG Dashboard

FG-RTN

Customer Return

VOL10 · MNT

Maintenance 설비관리

9 screens · L2 →

MNT-01

Equipment Card

MNT-02

Failure Register

MNT-03

OEE Analysis

MNT-04

Mold Management

MNT-05

PM Schedule

MNT-06

Downtime Log

MNT-07

Work Order

MNT-08

Spare Parts

MNT-09

MNT Dashboard

VOL11 · RPT

Reports 분석리포트

10 screens · L2 →

RPT-01

Daily Production

RPT-02

Defect Pareto

RPT-03

Daily Shipment

RPT-04

On-Time Delivery

RPT-05

Inventory Status

RPT-06

Equipment OEE

RPT-07

Monthly KPI

RPT-08

Schedule Adherence

RPT-09

Report Center

RPT-10

Report Builder

VOL12 · SYS

System Admin 시스템관리

8 screens · L2 →

SYS-01

User Management

SYS-02

RBAC Permissions

SYS-03

Factory Calendar

SYS-04

Interface Monitor

SYS-05

Audit Log

SYS-06

Notification Settings

SYS-07

System Configuration

SYS-08

System Health

VOL13 · MD

Master Data 기준정보 · 29 엔티티

5 hubs + 29 entities · L2 →

CATEGORY L3 HUBS · 카테고리 허브

LS - A Foundation	LS - B Routing & Production	LS - C Resources & Equipment	LS - D RFID & Materials	LS - E Quality & Logistics
ENTITY TABLE-DESIGN PAGES - 엔티티 테이블 설계서				
MD-01 Item Master	MD-02 BOM	MD-03 BOM Version	MD-04 BOP Routing	MD-05 Work Center
MD-06 Inspection Std	MD-07 Vendor	MD-08 Equipment	MD-09 Mold	MD-10 Paint / Fabric
MD-11 Shipment Dest	MD-12 Customer	MD-13 UOM	MD-14 Calendar	MD-15 Jig
MD-16 RFID Tag	MD-17 RAL Paint Color	MD-18 Oven	MD-19 RFID Reader	MD-20 Line Master
MD-21 Defect Code	MD-22 Defect Cause	MD-23 Packaging Spec	MD-24 Label Template	MD-25 Reason Code
MD-26 Common Code Group	MD-27 Spare Parts	MD-28 PM Template	MD-29 Line Time Pattern	

REF

Data Model

데이터 모델 참조

2 references

ERD
Entity Relationship Diagram

DB
Database Schema

MODULE	RULE	규칙 요약	ACTION
WH	BR-WH-001	입고 수량 $\pm 5\%$ 초과 시 SCM 승인 요청	Block / Alert
WH	BR-WH-002	로케이션 용량 초과 시 입고 차단	Block
PP	BR-PP-001	부품 재고 부족 시 WO 차단 (MRP)	Block / Alert
PP	BR-PP-002	WO 확정 후 자재 자동 예약	Auto-Reserve
INJ	BR-INJ-001	불량률 $\geq 5\%$ 시 라인 자동 정지	Stop / Alert
INJ	BR-INJ-002	금형 쇼트수 90% → 경고, 100% → 강제차단	Warn / Block
IMG	BR-IMG-001	원단 색상 불일치 시 PDA 다음 단계 차단	Block
PNT	BR-PNT-001	5단계 공정 순서 강제 적용 (건너뛸 불가)	Block
PNT	BR-PNT-002	건조로 온도 이탈 시 Andon 발동	Alert / Andon
QC	BR-QC-001	3회 연속 불량 시 강화검사 플래그 자동 설정	Flag
QC	BR-QC-002	QC FAIL → 해당 WO 완료 차단	Block
FG	BR-FG-001	FIFO 원칙: 입고일 오래된 로트 우선 출고	Sort / Alert
MNT	BR-MNT-001	OEE $< 65\%$ 시 설비 점검 알림	Alert
SYS	BR-SYS-001	로그인 5회 실패 시 계정 잠금	Lock
MD	BR-MD-001	Phase-0 미완료 품목은 WO 생성 차단	Block
MD	BR-MD-33	라인 시간패턴 세그먼트는 0~1440분을 빈틈·중첩 없이 분할	Block
PP	BR-PP-031	WO 블록은 단일 OPERATING 가동창 안에 완전히 배치	Block
PP	BR-PP-035	계획 가동창 내 설비 정지는 실제 비가동(UNPLANNED) 자동 분류	Auto
PP	BR-PP-042	OEE = 가용성 $\times$ 성능 $\times$ 품질, 기간별 스냅샷 저장	Auto

FROM	TO	DATA / EVENT	설명
SCM (ERP)	WH	PO Sync	발주 정보 자동 수신
SCM (ERP)	PP	Customer Order	수주 수신 → 생산계획 트리거
WH	PP	Stock Level	재고 수준 실시간 공유
PP	INJ / IMG / PNT	Work Order	WO 전달 → POP 화면 표시
INJ / IMG / PNT	QC	Production Result	생산 완료 → QC 검사 트리거
QC	FG	PASS Flag	합격 제품 → 완제품 창고 이동
FG	SCH (ERP)	Shipment Confirm	출하 완료 → ERP 전송
INJ	MNT	Mold Shot Count	금형 쇼트수 → 설비관리 모니터링
All	RPT	Transaction Data	전 모듈 데이터 → 리포트 집계
MD	ALL	Master Data	기준정보 → 전 모듈에 배포
MD-29	PP	Line Time Pattern	라인별 분단위 시간패턴 → 스케줄·가동 기준
PP-ODM	RPT	Operating State Log	가동·비가동 상태 → OEE 가용성 집계

DATA MODEL REFERENCES 데이터 모델 참조

ERD

ERD

Entity Relationship Diagram

전체 엔티티 관계도

Cross-module table relationships

전 모듈 테이블 관계 다이어그램

DB

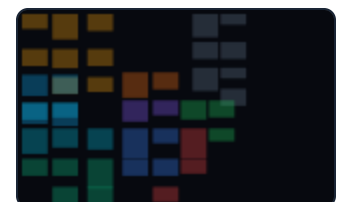
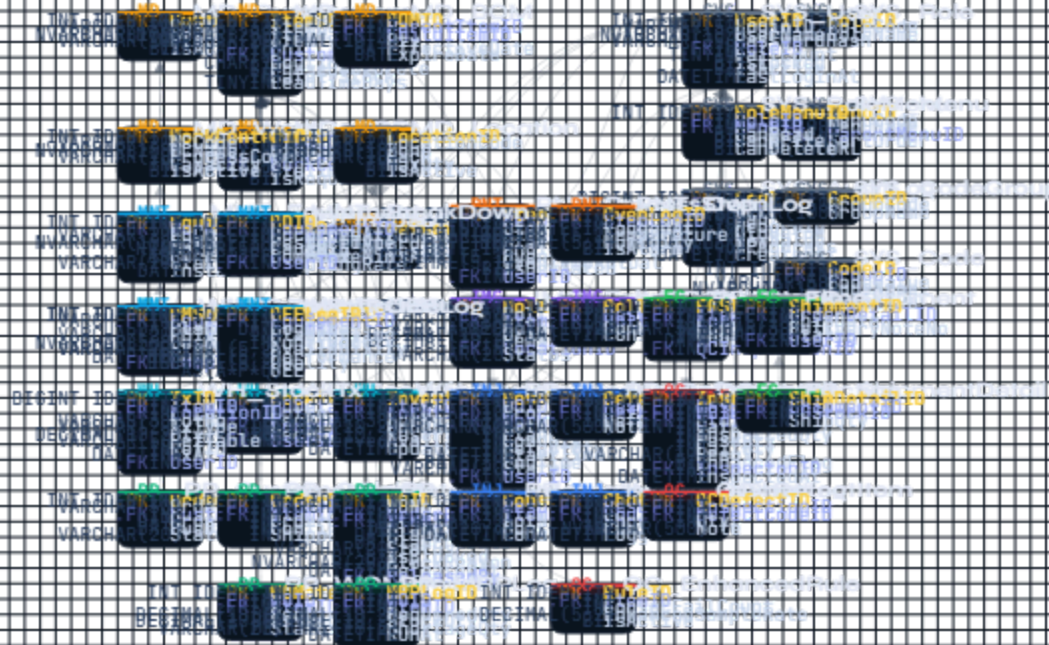
SCHEMA

Database Schema

데이터베이스 스키마

Full table / column DDL definitions

전체 테이블·컬럼 DDL 정의



A-MES Database Schema

MSSQL · .NET · 자동차 부품 MES · 간략 설계도

PK FK 클릭: 모듈 필터

ALL MD 기준정보 WH 창고 PP 생산계획 INJ 사출 IMG 감싸기 PNT 도장 QC 품질
FG 완제품 MNT 설비 SYS 시스템

MD — MASTER DATA 기준정보

MDMD_Customer거래선 PK CustomerIDINT IDENTITY CustomerCodeVARCHAR(20)SAV / GEO CustomerNameNVARCHAR(100) PlantVARCHAR(10)Savannah/G... IsActiveBITDEFAULT 1
--

WH — WAREHOUSE 창고관리

WHWH_PurchaseOrder발주(SCM)	WHWH_Receiving입고
--------------------------------------	-----------------------------

PK POID	INT IDENTITY	PK ReceivingID	INT IDENTITY
PONo	VARCHAR(30) SCM PO번호	FK POID	INT → WH_Purc...
FK CustomerID	INT → MD_Cust...	FK ItemID	INT → MD_Item
PODate	DATE	LotNo	VARCHAR(30)
Status	VARCHAR(20) Pending/Re...	ReceivedQty	DECIMAL(10,2)
		FK LocationID	INT → MD_Loca...
		ReceivedAt	DATETIME
		FK UserID	INT → SYS_User

WH WH_Inventory	재고 (현재고)	
PK InventoryID	INT IDENTITY	
FK ItemID	INT → MD_Item	
FK LocationID	INT → MD_Loca...	
LotNo	VARCHAR(30)	
OnHandQty	DECIMAL(10,2)	
ReservedQty	DECIMAL(10,2) WO 예약량	
UpdatedAt	DATETIME	

● PP – PRODUCTION PLAN

생산계획

PP PP_CustomerOrder	수주	PP PP_WorkOrder	작업지시
PK OrderID	INT IDENTITY	PK WOID	INT IDENTITY
OrderNo	VARCHAR(30) SCM 수주번호	WONo	VARCHAR(30) UNIQUE
FK CustomerID	INT → MD_Cust...	FK OrderID	INT → PP_Cust...
FK ItemID	INT → MD_Item	FK ItemID	INT → MD_Item
OrderQty	INT	PlanQty	INT
DueDate	DATE	PlanStartDate	DATE
Status	VARCHAR(20) Open/Closed	Status	VARCHAR(20) Draft/Rele...
		IsBlocked	BIT MRP 차단

● INJ / IMG / PNT – PRODUCTION

생산실적 (공통구조)

PROD PR_ProductionResult	생산실적	PROD PR_DefectDetail	불량 상세
PK ResultID	INT IDENTITY	PK DefectID	INT IDENTITY
FK WOID	INT → PP_Work...	FK ResultID	INT → PR_Prod...
ProcessCode	VARCHAR(10) INJ / IMG ...	DefectCode	VARCHAR(20) INJ-D01 등
FK WorkCenterID	INT → MD_Work...	DefectQty	INT
GoodQty	INT 양품	Note	NVARCHAR(500)
DefectQty	INT 불량		
StartTime	DATETIME		
EndTime	DATETIME		

FK UserID		INT → SYS_User	
PNT	PNT_ProcessStep	도장 단계 이력	
PK	StepLogID	INT IDENTITY	
FK	ResultID	INT → PR_Prod...	
	StepSeq	TINYINT	1~5
	StepName	VARCHAR(30)	탈지/하도/...
	CompletedAt	DATETIME	
	OvenTemp	DECIMAL(5,1)	건조로 온도...
● QC – QUALITY CONTROL 품질관리			
QC	QC_Inspection	검사 헤더	
PK	InspectionID	INT IDENTITY	
FK	ResultID	INT → PR_Prod...	
FK	WOID	INT → PP_Work...	
	Gate	TINYINT	1 / 2 / 3
	InspectedQty	INT	
	PassQty	INT	
	FailQty	INT	
	Result	VARCHAR(10)	PASS / FAIL
	EnhancedFlag	BIT	강화검사 여부
FK	InspectorID	INT → SYS_User	
	InspectedAt	DATETIME	
● FG – FINISHED GOODS 완제품출고			
FG	FG_Stock	완제품 재고	
PK	FGStockID	INT IDENTITY	
FK	ItemID	INT → MD_Item	
FK	LocationID	INT → MD_Loca...	
	LotNo	VARCHAR(30)	
	OnHandQty	INT	
	InboundDate	DATE	FIFO 기준
FG	FG_Shipment	출하	
PK	ShipmentID	INT IDENTITY	
FK	OrderID	INT → PP_Cust...	
FK	FGStockID	INT → FG_Stock	
	ShipQty	INT	
	ShipDate	DATETIME	
	DeliveryNote	VARCHAR(30)	거래명세서 ...
FK	UserID	INT → SYS_User	
● MNT – MAINTENANCE 설비관리			
MNT	MNT_Equipment	설비 기준	
PK	EquipmentID	INT IDENTITY	
FK	WorkCenterID	INT → MD_Work...	
MNT	MNT_PMSchedule	PM 계획	
PK	PMScheduleID	INT IDENTITY	
FK	EquipmentID	INT → MNT_Equ...	

EqCode	VARCHAR(20)		PlannedDate	DATE	
MoldShotLimit	INT	금형 한계 ...	PMTYPE	VARCHAR(20)	Daily/Week...
CurrentShotCount	INT	누적 쇼트수	Status	VARCHAR(20)	Plan/Done/...
Status	VARCHAR(20)	Active/PM/...			
MNT	MNT_OEELog	OEE 실적			
PK	OEELogID	INT IDENTITY			
FK	EquipmentID	INT → MNT_Equ...			
	LogDate	DATE			
	Availability	DECIMAL(5,2) %			
	Performance	DECIMAL(5,2) %			
	Quality	DECIMAL(5,2) %			
	OEE	AS (A*P*Q) Computed			
●	SYS – SYSTEM	시스템 / 공통			
SYS	SYS_User	사용자	SYS	SYS_Role	역할 (RBAC)
PK	UserID	INT IDENTITY	PK	RoleID	INT IDENTITY
	LoginID	VARCHAR(50) UNIQUE		RoleCode	VARCHAR(10) ADMIN/MGR/...
	PasswordHash	VARCHAR(256) BCrypt		RoleName	NVARCHAR(50)
FK	RoleID	INT → SYS_Role			
	FailCount	TINYINT 5회 잠금			
	IsLocked	BIT			
	IsActive	BIT			
SYS	SYS_AuditLog	감사로그			
PK	LogID	BIGINT IDENTITY			
FK	UserID	INT → SYS_User			
	Action	VARCHAR(50) INSERT/UPD...			
	TableName	VARCHAR(50)			
	RecordID	INT			
	OldValue	NVARCHAR(MAX) JSON			
	NewValue	NVARCHAR(MAX) JSON			
	CreatedAt	DATETIME 5년 보관			

총 24개 테이블 MD: 6 · WH: 3 · PP: 2 · PROD: 3 · QC: 1 · FG: 2 · MNT: 3 · SYS: 3 · PNT: 1

MSSQL 2019+ · .NET 8 EF Core